



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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ANISH NAIR

BATCH: A1

PRN:25070122037

Subject: Programming with Java

ASSIGNMENT No: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcome

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 1

Enter course name: Java Programming

Course registered.

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 1

Enter course name: Data Structures

Course registered.

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 3

---- Registered Courses ----

```

4  public class CourseRegistrationDemo {
5      public static void main(String[] args) {
9          do {
14             switch (choice) {
15                 case 1:
19                     break;
20                 case 2:
21                     System.out.print("Enter course name to remove: ");
22                     String courseToRemove = sc.nextLine();
23                     if (courses.remove(courseToRemove)) {
24                         System.out.println("Course removed.");
25                     } else {
26                         System.out.println("Course not found.");
27                     }
28                     break;
29                 case 3:
30                     StringBuffer sb = new StringBuffer();
31                     sb.append("---- Registered Courses ----\n");
32                     for (int i = 0; i < courses.size(); i++) {
33                         sb.append((i + 1) + ". " + courses.get(i) + "\n");
34                     }
35                     System.out.println(sb);
36                     break;
37                 case 4:
38                     System.out.println("Exiting...");
39                     break;
40                 default:
41                     System.out.println("Invalid choice.");
42             }
43         } while (choice != 4);
44     }
45 }

```

```

1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class CourseRegistrationDemo {
5      public static void main(String[] args) {
6          ArrayList<String> courses = new ArrayList<>();
7          Scanner sc = new Scanner(System.in);
8          int choice;
9          do {
10             System.out.println("\n1. Add Course\n2. Remove Course\n3. View Registered Courses");
11             System.out.print("Enter choice: ");
12             choice = Integer.parseInt(sc.nextLine());
13
14             switch (choice) {
15                 case 1:
16                     System.out.print("Enter course name: ");
17                     courses.add(sc.nextLine());
18                     System.out.println("Course registered.");
19                     break;
20                 case 2:
21                     System.out.print("Enter course name to remove: ");
22                     String courseToRemove = sc.nextLine();
23                     if (courses.remove(courseToRemove)) {
24                         System.out.println("Course removed.");
25                     } else {
26                         System.out.println("Course not found.");
27                     }
28                     break;
29                 case 3:
30                     StringBuffer sb = new StringBuffer();
31                     sb.append("---- Registered Courses ----\n");
32                     for (int i = 0; i < courses.size(); i++) {
33                         sb.append((i + 1) + ". " + courses.get(i) + "\n");
34                     }
35                     System.out.println(sb);
36                     break;

```

2. Finish assignment

1. Add Task

2. Remove Task

3. View Tasks

4. Exit

Enter choice: 2

Enter task to remove: Buy groceries

Task removed.

1. Add Task

2. Remove Task

3. View Tasks

4. Exit

Enter choice: 3

---- To-Do List ----

1. Finish assignment

1. Add Task

2. Remove Task

3. View Tasks

4. Exit

Enter choice: 4

Exiting...

1. Add Task
2. Remove Task
3. View Tasks
4. Exit

Enter choice: 1

Enter task: Buy groceries

Task added.

1. Add Task
2. Remove Task
3. View Tasks
4. Exit

Enter choice: 1

Enter task: Finish assignment

Task added.

1. Add Task
2. Remove Task
3. View Tasks
4. Exit

Enter choice: 3

---- To-Do List ----

1. Buy groceries

```

4   public class ToDoListDemo {
5       public static void main(String[] args) {
10          do {
15              switch (choice) {
16                  case 1:
19                      System.out.println("Task added.");
20                      break;
21                  case 2:
22                      System.out.print("Enter task to remove: ");
23                      String taskToRemove = sc.nextLine();
24                      if (tasks.remove(taskToRemove)) {
25                          System.out.println("Task removed.");
26                      } else {
27                          System.out.println("Task not found.");
28                      }
29                      break;
30                  case 3:
31                      StringBuffer sb = new StringBuffer();
32                      sb.append("---- To-Do List ----\n");
33                      for (int i = 0; i < tasks.size(); i++) {
34                          sb.append((i + 1) + ". " + tasks.get(i) + "\n");
35                      }
36                      System.out.println(sb);
37                      break;
38                  case 4:
39                      System.out.println("Exiting...");
40                      break;
41                  default:
42                      System.out.println("Invalid choice.");
43              }
44          } while (choice != 4);
45      }
46  }

```

```

1  import java.util.ArrayList;
2  import java.util.Scanner;
3
4  public class ToDoListDemo {
5      public static void main(String[] args) {
6          ArrayList<String> tasks = new ArrayList<>();
7          Scanner sc = new Scanner(System.in);
8          int choice;
9
10         do {
11             System.out.println("\n1. Add Task\n2. Remove Task\n3. View Tasks\n4. Exit");
12             System.out.print("Enter choice: ");
13             choice = Integer.parseInt(sc.nextLine());
14
15             switch (choice) {
16                 case 1:
17                     System.out.print("Enter task: ");
18                     tasks.add(sc.nextLine());
19                     System.out.println("Task added.");
20                     break;
21                 case 2:
22                     System.out.print("Enter task to remove: ");
23                     String taskToRemove = sc.nextLine();
24                     if (tasks.remove(taskToRemove)) {
25                         System.out.println("Task removed.");
26                     } else {
27                         System.out.println("Task not found.");
28                     }
29                     break;
30                 case 3:
31                     StringBuffer sb = new StringBuffer();
32                     sb.append("---- To-Do List ----\n");
33                     for (int i = 0; i < tasks.size(); i++) {
34                         sb.append((i + 1) + ". " + tasks.get(i) + "\n");
35                     }
36                     System.out.println(sb);

```


1. Java Programming
2. Data Structures

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 2

Enter course name to remove: Java Programming
Course removed.

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 3

---- Registered Courses ----

1. Data Structures

1. Add Course
2. Remove Course
3. View Registered Courses
4. Exit

Enter choice: 4

Exiting...



Conclusion : ArrayList provides dynamic, resizable storage for tasks/courses, while StringBuffer efficiently builds and displays formatted string output without creating multiple intermediate String objects

Exercise

1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.
2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the **registered** course list. The application should allow users to add, remove, and view registered courses.